Making Organizational Behavior Observable:
An Auditable Multimodal Infrastructure for Measuring Behavior from Audio and Video

Abstract

Recordings of organizational interactions are increasingly common but remain difficult to use as scientific data. Transcripts leave out vocal delivery, conversational timing, visible behavior, and production context. Audio and video retain this information, yet researchers still need to organize it into measures they can trace to the recording. We develop an open-source, AI-assisted measurement infrastructure that converts heterogeneous recordings into time-aligned, source-linked evidence and enables researchers to construct and inspect multimodal measures across settings. We demonstrate the infrastructure using audio from 286 earnings calls involving 12 executives and video from 30 public interviews. The pilots show that routine language can be paired with unusual vocal delivery or timing, and that words, voice, and visible behavior can follow patterns not captured in transcripts alone. AI helps extract and organize observations at scale; researchers decide what to measure, how to validate it, and what conclusions the evidence supports. By linking measures to the original recordings and documenting how they were produced and where evidence is uncertain or missing, the system helps researchers check, revise, and adapt measures across settings.

1. The problem: From recorded to research-observable behavior

Organizational and social scientists study how people communicate, influence one another, and coordinate action. These processes involve words, vocal delivery, timing, visible behavior, and responses from others (LeBaron et al., 2018; Lehmann-Willenbrock, 2025). Researchers often rely on transcripts, surveys, or downstream outcomes because extracting and aligning this information from recordings is costly. Even studies that use recordings often compress moment-to-moment interaction into a few static variables. As a result, testing theories of organizational behavior across large numbers of naturally occurring interactions remains difficult.

This difficulty arises for multiple reasons. First, when information from one or more channels is discarded, we encounter *channel loss*. For example, a transcript preserves words but not how they were spoken, when a response began, whether speakers overlapped, or how others visibly reacted. Second, *measurement friction* arises when behavior can be heard or seen but is too costly to structure and compare across many recordings, such as capturing synchronicity of movements between speakers, or changes in speech pacing and pitch. Third, when the recording itself does not capture the relevant behavior, for instance because a listener was off-camera or a face was obscured, we encounter *source non-observability*. The first two issues restrict what researchers can feasibly measure, while the third issue characterizes the limits of what the available evidence can support.

Recent advances in acoustic analysis, speaker identification (diarization), speech recognition, language models, and computer vision can reduce channel loss and measurement friction. While they cannot reconstruct behavior that the source never captured, they can reliably document evidence that is unavailable or uncertain. Accurately detecting a pause, pitch change, facial movement, or camera cut clearly does not establish what that observation means theoretically (Russell et al., 2003; Barrett et al., 2019). Observable behaviors may have different meanings across people and situations, and model choices can alter extracted outputs and downstream conclusions (Harvey et al., 2026; Carlson & Burbano, 2026). The challenge is therefore not simply to extract more and more features from the raw data but to organize the extracted multimodal evidence in ways that clearly distinguishes observable from unobservable or uncertain information and maps them to construct definitions and theoretical interpretations.

2. Our contribution: An auditable multimodal measurement architecture

Prior research has developed key components of our measurement architecture, including theories and coding methods for speech, facial movement, gaze, turn-taking, interaction sequences, and automated tools that extract these features from raw data (Baltrušaitis et al., 2018; Liu et al., 2023). Multimodal corpora and studies of teams and earnings calls also demonstrate the value of combining language, voice, visible behavior, and context (Reece et al., 2022; Lehmann-Willenbrock & Hung, 2024; Mayew & Venkatachalam, 2012; Rennekamp et al., 2022; Baik et al., 2025). However, the field needs an auditable multimodal measurement architecture, which combines outputs from separate extraction tools into measures, linked to the underlying evidence, that can be inspected, updated, validated, and tested across settings (Figure 1).

The *evidence layer* addresses measurement friction by turning outputs from heterogeneous recordings into a common structure that can be compared across people and interactions. It aligns observations by participant, time, and channel and records the source, whether the relevant information was available, any quality concerns, and the model and version that produced it. This reduces the cost of locating, structuring, and aligning behavior across large numbers of recordings. It also makes limits such as non-observability explicit. For example, an off-camera participant is not coded as showing no facial movement, and an uncertain speaker assignment is treated as unidentifiable. Aligning observations across channels also reduces channel loss by preserving information and interaction sequences that

would disappear if recordings were reduced to transcripts or isolated features (Knox & Lucas, 2021).

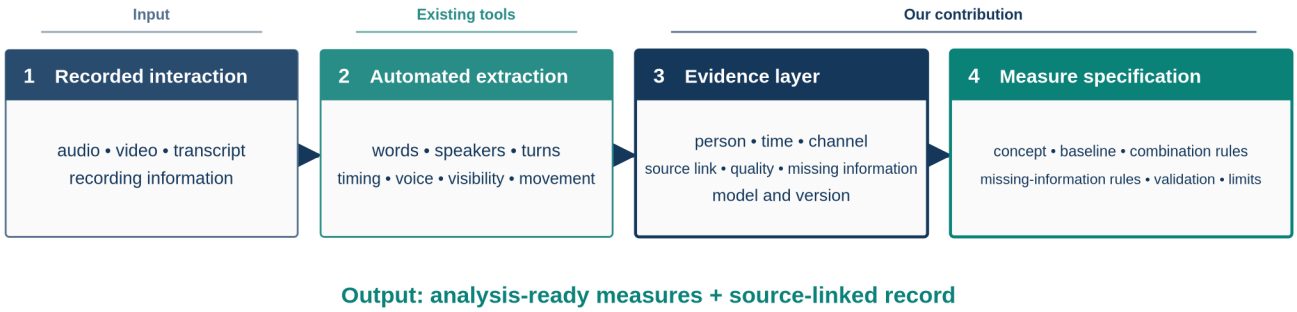


Figure 1. Proposed measurement architecture. *Multimodal raw data is processed using a suite of available tools to extract information. The proposed architecture aligns their outputs by person, time, and channel and links them to the source. Researchers specify how to combine observations into measures for analysis. Extraction, measurement, and interpretation remain separate.*

The *measure-specification layer* addresses a second form of measurement friction by reducing the work required to turn structured behavioral evidence into analysis-ready variables. Researchers specify the construct, included observations, comparison baseline, treatment of unavailable information, aggregation rules, and validation plan. This makes measurement choices explicit and allows researchers to revise them and compare alternative specifications without rebuilding the underlying evidence. A pause, for example, can be recorded as a duration and as a departure from a stated baseline, but it is not automatically labeled as uncertainty. Researchers remain responsible for deciding whether and how such observations represent a theoretical construct. A short narrated example in the repository follows one executive answer from the source audio and transcript through its component measurements, showing how the evidence behind a measure can be inspected directly.[1]

3. Prototype and initial evidence: two pilot applications

We tested a working prototype of the proposed architecture in two settings with different data limitations. The earnings-call pilot asks whether audio can reveal departures from usual behavior that transcripts miss. The video-interview pilot examines how camera choices affect which behaviors can be observed. The pilots demonstrate the architecture's capabilities, without establishing that the observed departures measure any particular theoretical construct.

Earnings-call pilot. We collected 300 calls involving 12 executives across seven sectors. The 286 calls that met the analysis requirements contained approximately 313 hours of audio and 3,273 executive answers during live Q&A. Using each executive's answers across calls, we established a personal baseline and compared each answer with their usual language, vocal delivery, and interaction timing. In 177 answers, language was relatively typical for the executive, while vocal delivery or interaction timing was unusual (Figure 2). These answers had a multimodal departure score of at least 2.0 and a transcript departure score of no more than 1.25. The results do not automatically suggest intent or emotions of the speaker, but they warrant further investigation. The system retains continuous scores so researchers can inspect the size and source of each departure.

We compared our measure of vocal delivery departure to a transcript-based one to ask whether audio and interaction features flag answers that simple transcript features would miss because features like answer length, filled pauses, discourse fillers, and adjacent repetitions are not observed in transcripts. Using the same departure-score cutoff of 2.0, the audio/interaction measure flagged 325 answers and the transcript

[1] The project repository contains the measurement-specification prototype, derived data, figures, technical appendix, and researcher guide: https://github.com/heeyonkim/auditable-multimodal-measurement.

measure flagged 321, but only 66 answers appeared in both groups. In other words, 80% of the answers flagged by the audio/interaction measure were not flagged by the transcript measure. This does not establish which measure is better. It shows that audio and interaction features direct researchers toward a substantially different set of answers to inspect.

To further explore whether these departures might relate to organizational outcomes, we linked call-level scores to market reactions. Across 274 calls with sufficient eligible Q&A answers, greater median departure from an executive's usual vocal delivery and interaction timing was predictive of slightly lower market-adjusted returns, after accounting for transcript-visible departure, firm, and year (partial Spearman correlation = −.10, p = .094). This exploratory analysis demonstrates how the measures can be used in a study; it does not yet establish a causal relationship.

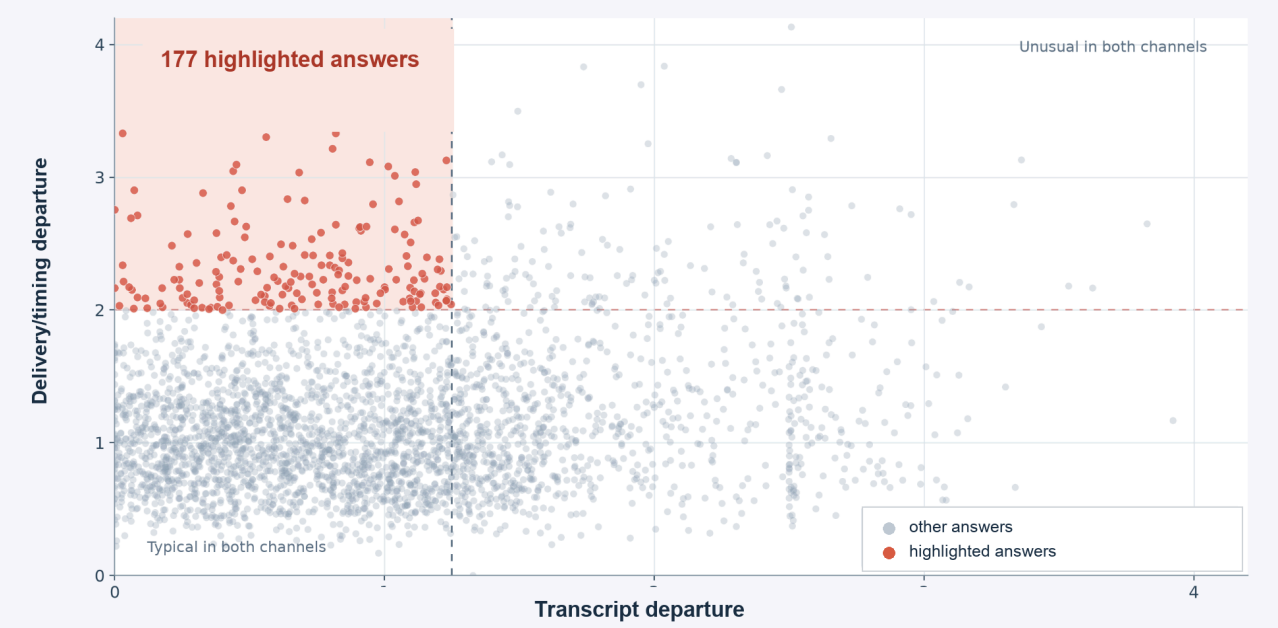


Figure 2. Audio surfaces departures that transcripts miss. *Each point represents an answer compared with that executive's own baseline. Highlighted answers were relatively typical in transcript language but unusual in vocal delivery or interaction timing.*

Video-interview pilot. Video captures visible behavior missing from transcripts and audio, but analyzing it systematically at scale remains difficult. Researchers need to align visible behavior with speech and interaction timing while accounting for camera selection and editing (LeBaron et al., 2018; Lehmann-Willenbrock & Hung, 2024). We examined 30 Bloomberg interviews spanning 8.6 hours and several production formats. In a typical interview, about two-thirds of speaker changes happened within two seconds before or after a camera cut, compared with about one-sixth expected by chance.

Ignoring editing could lead a researcher to mistake a production decision for a change in participant behavior. The prototype automatically detects camera cuts and records participant visibility separately. When someone is off-camera, their visible behavior is treated as unavailable rather than as an absence of facial movement.

We also examined eight interviews from Bloomberg's Leaders with Lacqua, comparing interviewer Francine Lacqua's transcript features, turn-taking, vocal delivery, and visible speaking and listening behavior across encounters with different CEOs. Each encounter was compared with other eligible interviews. In her interview with CEO Federico Marchetti, transcript features were close to her usual

pattern, while turn-taking and vocal delivery differed (Figure 3).

A sensitivity analysis illustrated the consequences of how unavailable visual information is handled. Treating periods when Francine Lacqua was not measurably visible as showing no facial movement produced a different ranking from the production-aware approach, which treated those periods as unavailable and accounted for framing and camera cuts. Six interviews were eligible under both specifications. Among them, the interview with CEO Pascal Soriot moved from fifth to first in measured distinctiveness when unavailable visual information, framing, and camera cuts were handled explicitly.

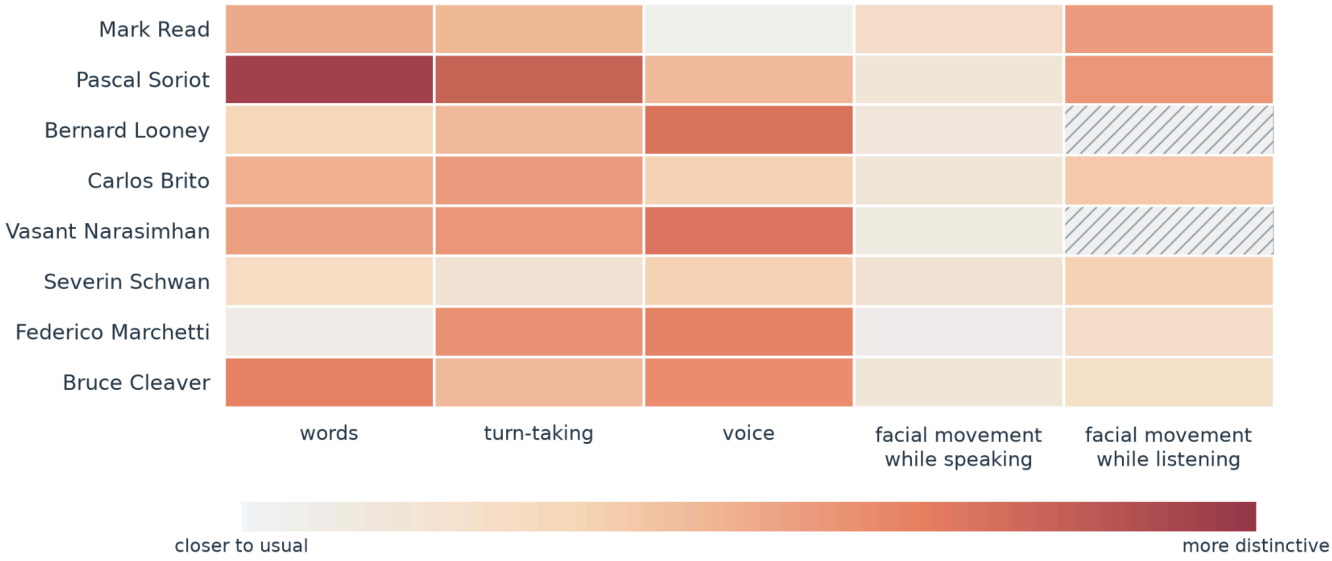


Figure 3. Behavioral patterns vary across channels. *Rows represent interviews, labeled by the CEO interviewed, and columns represent channels. Color shows how far the interviewer departed from her pattern in other eligible interviews. There were seven comparison interviews for words, turn-taking, voice, and facial movement while speaking, and five for facial movement while listening. Hatched cells indicate insufficient observable evidence to calculate a score.*

Together, the pilots show that the system can recover information omitted from transcripts, compare behavior within people, and distinguish unavailable visual information from an absence of behavior.

4. Stage 2: From working prototype to reusable measurement infrastructure

In Stage 2, we plan to extend and validate the prototype across a wider range of recordings and research settings. The work will cover technical accuracy, measure validation, researcher use, and empirical expansion.

Technical benchmarking. We will check the infrastructure's identification of speakers, turns, timing, vocal features, camera cuts, visibility, and facial movement against human-reviewed examples for datasets that are commonly used in organizational behavior. Tests will cover overlapping speech, poor audio, network delay, obscured faces, and rapid editing. We will report accuracy, unavailable information, required corrections, processing time and cost, and differences across recording conditions.

Measure validation. We will evaluate proposed measures separately from extraction accuracy, since correctly detecting a pause, pitch change, or smile does not establish its meaning. For each measure, researchers will specify the construct, observations, baseline, combination rule, and treatment of unavailable information. We will assess agreement with human coding or established measures, stability under reasonable measurement choices, information added beyond transcripts, and performance in new

settings. We will retain null findings and disagreements across channels.

Researcher use. A guided interface will allow researchers to process recordings, inspect the source of an observation, correct assignments, specify and compare measures, and export results. Researchers outside the project will reproduce the pilots and develop new measures so we can assess whether the infrastructure is extensible and can be used independently.

Empirical expansion. We will enlarge the earnings-call executive panel, construct personal baselines using only earlier observations, and define outcome tests in advance. For video, we will study repeated interviews with the same executives and interviewers across programs and over time. These studies will help establish when particular measures are informative and where their limits lie.

In sum, Stage 2 will deliver a documented infrastructure released under appropriate open-source terms, a versioned framework for defining measures, a human-validated benchmark set, and a technical appendix reporting performance and limitations.

5. Risks and boundary conditions

The main risks concern false precision, unsupported psychological interpretation, demographic and cultural bias, privacy, and changes in the models used. Microphones, network delay, camera editing, and recording settings can also introduce differences. We will address these risks by linking observations to the original recording, reporting unavailable information and exclusions, and comparing reasonable models and measurement choices. Where appropriate, we will use comparisons within people or production formats. We will report differences across people and recording conditions rather than average them away.

The infrastructure will not infer emotion, deception, intent, personality, culture, or executive quality from behavioral information without a theory and independent evidence supporting the interpretation. Even accurately measured pauses or facial movements may not make a construct observable from a recording. Establishing these limits will help prevent unsupported claims.

6. How organizational and social science would change

If successful, the project would allow researchers to observe more of the behavior captured in organizational recordings. Material now reduced to transcripts or left unused could support studies of how people speak, respond, listen, influence one another, coordinate, and change across encounters. Researchers could examine moment-to-moment interaction across available recordings at scale, extending work that currently depends on small, manually coded samples. Additionally, each measure will remain linked to its underlying observations and the original recording, allowing other researchers to inspect, revise, test, and adapt it.

In conclusion, the aim is neither faster transcription nor automated theoretical labeling. AI extracts and organizes information at scale, while researchers define constructs, choose measures, validate interpretations, and remain responsible for inference. Reducing the cost of structuring recordings while preserving the source evidence would allow researchers to study questions about organizational interaction that are currently infeasible.

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